



中药材桑白皮、构树根皮药效差异的物质基础

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摘要: **目的** 探讨传统中药材桑白皮和构树根皮药效差异, 解析桑白皮与构树根皮中代谢物的差异。 **方法** 试验采用超高效液相色谱串联质谱 (UPLC-MS/MS) 技术进行广泛靶向代谢组学分析, 采用正交偏最小二乘判别分析 (OPLS-DA) 筛选差异代谢物。 **结果** 共鉴定到 575 个代谢物, 检测数据和主成分分析 (PCA) 显示, 桑树品种之间的共有代谢物数量明显多于桑树与构树间的共有代谢物数量, 桑白皮与构树根皮间的代谢物组成差异较大。共筛选出差异显著代谢物 489 个, 这些代谢物在桑树与构树间的差异大于桑树品种间的差异, 其主要的差异成分为酚酸类和黄酮类物质。通过设置筛选标准, 最终发现有 73 种关键物质在桑树和构树间差异明显, 其中 2 种物质 (香草酰咖啡酰酒石酸和 1-O-水杨酰- β -D-葡萄糖) 为桑白皮特有成分。以构树根皮为对照, 73 种关键代谢物在两个桑树品种 (冀桑 3 号和丰驰桑) 中均呈现相同的表达趋势, 41 种代谢物上调表达, 32 种代谢物下调表达。进一步分析发现, 桑树与构树差异代谢物的主要区别在于桑树酚酸类物质多呈现下调表达趋势, 而黄酮类、鞣质类、萜类、木脂素和香豆素类以及生物碱类物质则以上调表达为主。通过对差异代谢物进行注释分析, 发现桑白皮的功能性成分种类多样, 主要以黄酮类为主, 其他类、酚酸类、木脂素与香豆素类、生物碱类、萜类和 16 鞣质类物质均有展现; 而构树根皮则主要是酚酸类和黄酮类。因此, 功能性成分的偏好可能是桑白皮与构树根皮药效差异的主要原因。

关键词: 桑白皮; 构树根皮; 代谢组学; 差异代谢物

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The material basis of pharmacodynamic difference between root–root–bark mulberry and root–bark broussonetia

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Abstract: Objective To investigate the pharmacodynamic difference between mulberry root-bark and bark of traditional Chinese medicinal materials, and to analyze the difference of metabolites in mulberry bark and mulberry bark. Methods Ultra-high performance liquid chromatography tandem mass spectrometry (UPLC-MS/MS) was used to perform extensive targeted metabolomics analysis, and orthogonal partial least squares discriminant analysis (OPLS-DA) was used to screen differential metabolites. Results A total of 575 metabolites were identified. The detection data and principal component analysis (PCA) showed that the number of common metabolites between mulberry cultivars was significantly higher than that between mulberry and conformation tree, and the metabolites between mulberry white bark and conformation tree root bark were significantly different. A total of 489 metabolites with significant differences were screened, and the differences between mulberry and conformation trees were greater than those between mulberry varieties. The main components of these metabolites were phenolic acids and flavonoids. By setting the screening criteria, 73 key substances were found to be significantly different between mulberry and conformation trees, among which 2 substances (vanillyl cafeoyl tartaric acid and 1-O-salicyl- β -D-glucose) were unique components of mulberry white bark. The expression of 73 key metabolites showed the same trend in two mulberry varieties (Jisang No. 3 and Fengqisang). 41 metabolites were up-regulated and 32 metabolites were down-regulated. Further analysis showed that the main difference between the different metabolites of mulberry and conformation tree was that phenolic acids showed down-regulated expression, while flavonoids, tannins, terpenoids, lignans, coumarins and alkaloids were mainly regulated expression. Through annotation analysis of the differential metabolites, it was found that the functional components of mulberry bark were diverse, mainly flavonoids, other substances, phenolic acids, lignans and coumarins, alkaloids, terpenoids and tannins were shown. The root bark of the conformation tree is mainly phenolic acids and flavonoids. Therefore, the preference of functional components may be the main reason for the difference in efficacy between mulberry root-bark and mulberry bark.

Keywords : Mulberry root-bark; Broussonetia plants root-bark; Metabolomics; Differential metabolites